



Spirit of India Special

IGNITED

20 INNOVATORS WHO A

| By Kaveree Bamzai |

Sheikh Jehangir cannot read or write, hasn't been trained in any vocational skill and started working at the age of 11 to support his family. Yet, the 50-year-old car painter in Jalgaon, Maharashtra, has a patent to his name, and has applied for a second one. His latest innovation, a scooter-powered flourmill created to help his wife grind wheat even during long power cuts, inspired an invention featured in the monster hit *3 Idiots*. Jehangir's shop was removed from the roadside in a cleanliness drive and even though he was promised a new one by the government, he has not received it yet. What's more, when Jehangir applied to a public sector for a loan to support his innovation, he was refused one without a collateral of Rs 15 lakh.

Is it any wonder then, that the Economist Intelligence Unit's 2007 ranking placed India 57th and China 58th on the innovation scale but predicted, that at the present rate, by 2012, China would be 50th while India would move up only one rank. China, in particular, has overtaken India in technology intensive manufacturing that includes high-tech semiconductors and microchips to become the largest producer of electronics in the world. What's more, according to data from a US thinktank, Battelle Institute, India spends 0.9 per cent of GDP on R&D while China spends 1.5 per cent.

As IIM-Bangalore professor and author of *From Jugaad to Systematic Innovation*, Rishikesh T. Krishnan points out, innovation in pre-liberalisation India was aimed at three things: import substitution, making use of local resources (such as Amul making milk powder from buffalo milk) and lowering costs. Post-1991, there have been innovations in various areas, especially transport and pharma, but not enough. Even though the Big Three IT giants—Infosys, TCS and Wipro—have developed the global delivery model that is a breakthrough in software solutions, India needs more innovation, especially in education and healthcare, where the challenges are to reduce costs and increase access. But there are several factors holding it back, ranging from a culture where risk-taking is not tolerated to an education system that tends to encourage rote learning. Yes, the Government has made efforts to support innovation, be it through the National Innovation Foundation (NIF) or the Technopreneur Promotion Programme. But their scale needs to be upped. Remember Leonardo da Vinci flourished as an inven-

